PAN-ATLANTIC UNIVERSITY

DEPARTMENT OF ECONOMICS
SCHOOL OF MANAGEMENT AND SOCIAL SCIENCES
PAN-ATLANTIC UNIVERSITY, IBEJU-LEKKI, LAGOS
2025/2026 Academic Session (1st Semester Examination)

ECO 411: ADVANCED MICROECONOMICS **Number of Units: 3**

Instruction: Students are to attempt QUESTION ONE and any THREE other questions.

Time Allowed: 3 Hours

Question One **[2 marks Each = 20 marks]**

(a) A firm has a cost function given by $c(y) = 10y^2 + 1000$. What is its supply curve?

(b) A firm has a cost function given by $c(y) = 10y^2 + 1000$. At what output is average cost minimized?

(c) If the supply curve is given by $S(p) = 100+20p$, what is the formula for the inverse supply curve?

(d) A firm has a supply function given by $S(p) = 4p$. Its fixed costs are 100. If the price changes from 10 to 20, what is the change in its profits?

(e) If the long-run cost function is $c(y) = y^2 + 1$, what is the long-run supply curve of the firm?

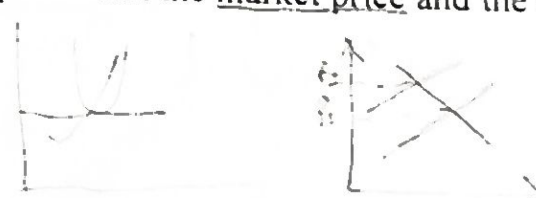
(f) What is the major assumption that characterizes a purely competitive market?

(g) In a purely competitive market a firm's marginal revenue is always equal to what? A profit-maximizing firm in such a market will operate at what level of output?

(h) If average variable costs exceed the market price, what level of output should the firm produce? What if there are no fixed costs?

(i) Is it ever better for a perfectly competitive firm to produce output even though it is losing money? If so, when?

(j) In a perfectly competitive market what is the relationship between the market price and the cost of production for all firms in the industry?



Question Two Done

Using the variable cost function of the form $c_v(y)$ and differential calculus, prove algebraically the following:

(a) If we are in a region where average variable costs (AVC) are falling, then it must be the case that the marginal costs (MC) are less than the average variable costs because it is the lower marginal costs that are pushing the average down. **(5 marks)**

(b) If we are in a region where the AVCs are minimised, then it must be the case that the MCs and AVCs are the same. **(5 marks)**

(c) If we are in a region where the AVCs are rising, then it must be that the MCs are higher than the AVCs because it is the higher MCs that are raising the average. **(5 marks)**

Question Three

(a) If the firm's production function exhibits constant returns to scale, it must always make zero long-run profits in equilibrium. Explain the possible reasons for this. **(6 marks)**

(b) Compare and contrast the factor demand curve and the inverse factor demand curve using a graphical aid. **(9 marks)**

Question Four

(a) If allowing a natural monopolist to set the monopoly price is undesirable due to the Pareto inefficiency and forcing the natural monopoly to produce at a competitive price is infeasible due to negative profits, discuss, using a well-labelled diagram, the possible options and problems that the country's authorities will face through strict regulations. **(6 marks)**

(b) Consider the short- and long-run effect of taxation on an industry with free entry and exit. Use appropriate aids to discuss your opinion. **(6 marks)**

(c) Use the marginal revenue and elasticity relationship to explain why a monopolist will never operate where the demand curve is *inelastic*. **(3 marks)**

Question Five

(a) Discuss the distinction between fixed costs and quasi-fixed costs. **(3 marks)**

(b) (i) With concrete illustrations, extensively discuss sunk costs. **(4 marks)**

(ii) While recoverable and sunk costs are very similar, their differences can be quite significant. Use concrete example to explain your position and what can be done to keep the distinction between both clear. **(2 marks)**

(c) (i) Given that a firm has the cost function of this nature,

$$c(w_1, w_2, y) = \left[\left(\frac{a}{b}\right)^{b/a+b} + \left(\frac{a}{b}\right)^{-a/a+b}\right] * w_1^{a/a+b} * w_2^{b/a+b} * y^{1/a+b}.$$

Explain the relationship of its cost with output if the firm's technology exhibits decreasing, constant, or increasing returns to scale depending on the value of $a + b$. **(4 marks)**

(ii) If the firm facing c(i) above is producing where $MP_1/w_1 > MP_2/w_2$, briefly explain what it can do to reduce costs but maintain the same output level? **(2 marks)**

Question Six

(a) Since the long-run effects of acquiring different fixed inputs and the entry and exit concepts are related, discuss their long run effects on industry equilibrium. **(5 marks)**

(b) Assume there are 1,...,4 firms operating in a market with identical long-run cost functions and free entry. Graph and explain the industry supply curves for the maximum number of firms that the market can accommodate. **(5 marks)**

(c) In the short run, with a fixed number of firms, an industry without entry barriers and zero profits has an upward-sloping supply curve. However, in the long run, with rising competitors, its supply curve is flat at the price equal to the minimum average cost. Use a graph to explain what will happen when the government imposes a tax on this industry. **(5 marks)**